

Riley Morris

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Technical Skills

Frontend Technologies: HTML, JavaScript, TypeScript, ReactJS, Tailwind CSS, Material-UI, Redux Toolkit, Jotai, React Query, Shadcn
Backend Technologies: Python, Node.js, NestJS, FastAPI, Prisma, Supabase, Elasticsearch, REST
Cloud & Infrastructure + Testing: AWS, Terraform, Jest, Playwright

Professional Experience

Software Engineer, Flock- Contract

Jan 2025 - Present

- Developed and optimized a referral management platform using React, Tailwind, and ShadCN, enhancing UI/UX with customizable and embeddable referral component
- Built and secured scalable backend APIs with NestJS and Supabase, enabling real-time analytics, third-party integrations and automated reward payouts, with Jest and Playwright ensuring high reliability through testing

Software Engineer, Sanofi

Oct 2022 - Dec 2024

- Designed and built a GenAI application to generate Clinical Study Reports (CSR) enabling medical writers to expedite drug-to-market timelines by 90%, built with React and NodeJS
- Developed reusable, responsive and accessible components and front-end library, built with React
- Led a significant refactor to modernize front-end codebases by integrating an in-house front-end library, reducing development time by 20% and improving maintainability and user experience
- Developed a full stack feature that enables data scientists and machine learning engineers to rapidly develop and deploy AI models and applications at scale, with React, Python and FastAPI
- Developed an internal search application, enabling 10,000+ employees the ability to find structured data (database tables, Excel files etc.) and unstructured data (PDFs, images etc.) significantly improving efficiency and reducing time spent on data retrieval
- Collaborated with product managers, engineers, and stakeholders to optimize the search engine by implementing a DisMax query in ES, enhancing recall by 15% without relevancy loss, while designing a scalable solution to handle high traffic and large datasets

Software Engineer Intern, Nference

Jan 2022 - Apr 2022

- Designed and implemented a React-based search interface, integrating NLP algorithms to identify medications in biomedical research papers, enhancing data accessibility and accelerating the discovery process for researchers
- Developed a new feature with a trie-based data structure to efficiently store and retrieve user input, enabling sub-millisecond query responses for datasets with over 1 million entries

Market Analyst Intern, TMX Group - Equity and Derivatives Trading Products

May 2021 - Aug 2021

- Provided quantitative and strategic support to foster TMX product growth, developing data-driven insights to optimize client interactions with Equity Trading products

Research Analyst Intern, Virtu Financial

May 2019 - May 2020

- Analyzed trading data for 90 institutional clients to identify execution trends, developing analytics reports and presentations on client-specific algorithmic performance for the Global Transaction Cost Analysis team

Education

Bachelor of Applied Science, University of Toronto

June 2022

Electrical & Computer Engineering, Minor in Business, Leadership & Communications Certificates

Software Engineering Projects

Relationship Management Library

- Designed and built a JavaScript library in vanillaJS for identifying and viewing the relationship between two nodes using unique data structures to represent the different connections
- Used Canvas API to visualize the graph structure on the DOM

Web-Based Navigation UI and Algorithm

- Led the research, design and development of an online navigation tool, a Google Map competitor, designed user interface for usability and responsiveness, and developed numerous performance-driven algorithms (Dijkstra) to optimize travel time